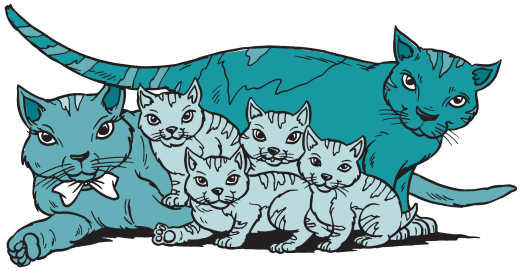
(ii) My family of 2 cats has increased 3 times. How many kittens do I have now?

$2 \times 3 = 6$

$6 - 2 = 4$

I have 4 kittens.



See decrease, progression

index

(Plural: indices)

10^{6} — index or exponent; base

Index is also called exponent.

In $\sqrt[3]{5}$ the index is 3. Where no index is written, as in $\sqrt{5}$, the index is 2.

See base, exponent, index notation, square root

index laws

In algebra, when working with indices or algebraic expressions, these laws must be remembered:

Law	*Example*
$x^a \times x^b = x^{a+b}$	$5^3 \times 5^2 = 5^{3+2} = 5^5$
$\frac{x^a}{x^b} = x^{a-b}$	$\frac{5^3}{5^2} = 5^{3-2} = 5^1 = 5$
$x^0 = 1$	$5^0 = 1$
$(x^a)^b = x^{a \times b} = x^{ab}$	$(5^3)^2 = 5^{3\times 2} = 5^6$
$(x \times y)^a = x^a y^a$	$(5 \times 4)^3 = 5^3 \times 4^3$
$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$	$\left(\frac{5}{4}\right)^3 = \frac{5^3}{4^3}$
$x^{-a} = \frac{1}{x^a}$	$5^{-3} = \frac{1}{5^3}$
$\sqrt[n]{a} = a^{\frac{1}{n}}$	$\sqrt[3]{5} = 5^{\frac{1}{3}}$

index notation

A shorthand way of writing large numbers such as 1 000 000. Also called scientific notation.

Example

Using index notation:

$$1\,000\,000 = 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 10^6$$

10^{6} — index or exponent; base

is read as:

'ten to the power of six' or

'ten to the sixth power'.

See base, cubed number, power of a number, scientific notation, square number